

Airbus Material Specification

Carbon Fiber Reinforced Epoxy Prepreg

Plain Weave Fabric / 180°C curing class - Flow controlled

Contents

1 Scope 3

2 Application 3

3 Normative References 3

4 Requirements..... 5

 4.1 General..... 5

 4.1.1 Material description 5

 4.1.2 Work- and storage life 5

 4.1.3 Processing 5

 4.2 Specific..... 6

 4.2.1 Physical-chemical properties 6

 4.2.2 Mechanical properties 7

5 Designation 12

6 Technical Specification 12

7 Incoming Inspection..... 12

Record of Revisions 13

AIRBUS | NormMaster

1 Scope

This Material Specification defines the requirements of carbon fiber epoxy prepreg, plain weave fabric; 180°C curing class, flow controlled for aerospace application.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures.

The requirements specified in this Material Specification shall not be used as stress allowable.

3 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ISO10119	Carbon fibres - Determination of density.
ISO1183	Plastics - Methods for determining the density and relative density of non-cellular plastics
EN2378	Aerospace series - Fibre reinforced plastics - Determination of water absorption by immersion
EN2557	Aerospace series - Carbon fibre preimpregnates - Determination of mass per unit area.
EN2558	Aerospace series - Carbon fibre preimpregnates - Determination of the percentage of volatile matter.
EN2559	Aerospace series - Carbon fibre preimpregnates - Determination of resin and fibre content and the mass of fibre per unit area.
EN2560	Aerospace series - Carbon fibre preimpregnates - Determination of the resin flow.
EN2561	Aerospace series - Unidirectional laminates carbon thermosetting resin - Tensile test parallel to the fibre direction.
EN2563	Aerospace series - Unidirectional laminates carbon thermosetting resin - Determination of interlaminar shear strength
EN2597	Aerospace series - Unidirectional laminates carbon thermosetting resin - Tensile test perpendicular to the fibre direction.
EN2743	Aerospace series - Standard procedure for conditioning prior to testing.
EN2823	Aerospace series - Fibre reinforced plastic - Determination of the effect of exposure to humid atmospheres on physical and mechanical characteristics
EN2850	Aerospace series - Carbon fibre thermosetting resin unidirectional laminates compression test parallel to the fibre direction
EN3615	Aerospace series - Fibre reinforced plastic - Determination of the conditions of exposure to humid atmosphere and moisture absorption.
ABS5346	Airbus Standard - Carbon fibre reinforced epoxy prepreg plain weave fabric / 180°C curing class - Flow Controlled

AIMS05-01-000 Part 1	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg - General requirements - Technical Specification
AIMS05-01-000 Part 6	Airbus Material Specification- Qualification Programme - Carbon fibre reinforced epoxy prepreg - Qualification program / batch release testing Fabric / 180°C - curing class - Flow controlled - Structural material - Technical Specification
AIMS05-01-000 Part 100	Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
AITM1-0001	Airbus Test Method - Determination of mechanical degradation due to chemical paint strippers
AITM1-0002	Airbus Test Method - Fibre reinforced plastics - Determination of in-plane shear properties ($\pm 45^\circ$ tensile test)
AITM1-0003	Airbus Test Method - Fibre reinforced plastic materials - Determination of the glass transition temperatures
AITM1-0005	Airbus Test Method - Carbon fibre reinforced plastics - Determination of interlaminar fracture toughness energy MODE I
AITM1-0006	Airbus Test Method - Carbon fibre reinforced plastics - Determination of interlaminar fracture toughness energy MODE II
AITM1-0007	Airbus Test Method - Fibre reinforced plastics - Determination of plain, open hole and filled hole tensile strength
AITM1-0008	Airbus Test Method - Fibre reinforced plastics - Determination of plain, open hole and filled hole compression strength
AITM1-0009	Airbus Test Method - Fibre reinforced plastics - Determination of bearing strength
AITM1-0010	Airbus Test Method - Fibre reinforced plastics - Determination of compression strength after impact
AITM3-0001	Airbus Test Method - Analysis of thermoset systems by high performance liquid chromatography
AITM3-0002	Airbus Test Method - Analysis of non-metallic materials (uncured) by differential scanning calorimetry
AITM3-0003	Airbus Test Method - Analysis of organic compounds by infrared spectroscopy
AITM3-0004	Airbus Test Method - Determination of the viscosity and gel time of thermosetting resin systems
AITM3-0008	Airbus Test Method - Determination of the extent of cure by differential scanning calorimetry
ASTM E831	Standard Test Method for Linear Thermal Expansion of Solid Materials by Thermo-mechanical Analysis

4 Requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material description

See Table 1.

Table 1: Material description

Characteristic		Requirements
Material type		Carbon fiber fabric/epoxy resin
Formulation	Fibre	Standard modulus carbon fibre
	Matrix	Epoxy, toughened, flow controlled
	Fabric	Plain weave fabric
Form and method of processing		Prepreg, cured in an autoclave at 180°C
Resin content by weight, uncured		44% ± 3%
Material code		As defined in Table 4
Service temperature range		-55°C up to 95°C

4.1.2 Work- and storage life

See Table 2.

Table 2: Worklife and storage conditions

Properties	Shop conditions	Unit	Requirements
Storage life (at -18°C)	-	months	> 6
Worklife ¹⁾ (Tacklife)	30°C / 75%RH	days	≥ 10
	(25 ± 2)°C / (45 ± 10)%RH	days	≥ 10
¹⁾ The prepreg can be exposed to an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use			

4.1.3 Processing

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3.

The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all the batch release test specimens.

Table 3: Processing conditions

Cure parameter	Unit	Requirements
Cure type: Autoclave		
Heating rate	°C/min	0,2 - 5,0
Cure temperature	°C	180 ^{+10/-5}
Cure time	min	120 - 240
Cure pressure	MPa	0,3 – 1,1
	bar	3 - 11
Cooling rate	°C/min	0,2 - 3,5
Vacuum (abs)	MPa	0,005 - 0,1
Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between manufacturer and purchaser.		

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see Table 4.

Table 4: Physical-chemical properties

Properties		Test method	Unit	Requirements ¹⁾
Material code				AIMS05-01-005A
Uncured				
Prepreg areal weight		EN2557C	g/m ²	345 ± 20
Fibre areal weight		EN2559C	g/m ²	193 ± 8
Resin density		ISO1183 Method A	g/cm ³	1,22 ± 0,03
Fibre density		ISO10119	g/cm ³	1,76 ± 0,04
Volatile content		EN2558	%	≤ 2
Resin content		EN2559	wt%	44 ± 3
Physical Chemical tests	HPLC	AITM3-0001		2)
	DSC	AITM3-0002		2)
	IR	AITM3-0003		2)
	Gel Time	AITM3-0004 method B or C		2)
Resin flow		EN2560	%	≥1
Tack		To be agreed between manufacturer and Airbus		-

(continued)

Table 4: Physical-chemical properties (concluded)

Properties	Test method	Unit	Requirements ¹⁾
Material code			AIMS05-01-005A
Cured			
CTE ³⁾	ASTM E831	°C-1	Results to be reported
Water uptake	EN3615, 70°C/85% R.H.	%	Results to be reported
Water uptake	EN2378, 70°C/water	%	Results to be reported
Extent of cure	AITM3-0008	%	≥ 90
Nominal cured ply thickness ⁴⁾	acc. to AIMS05-01-000 Part 1	mm	0,237

Notes:

¹⁾ The stated requirements are average values.

²⁾ Equipment settings and value to be agreed between manufacturer and Airbus, and recorded in the IPS.

³⁾ CTE is Coefficient of thermal Expansion.

⁴⁾ The nominal cured ply thickness is related to a fiber volume content of 46,3%.

4.2.2 Mechanical properties

See Table 5.

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning ²⁾	Test temp. [°C]	Requirements ¹⁾	
					X	Xmin
Material Code					AIMS05-01-005A	
Glass transition temperature, Tgonset	AITM1-0003	°C	dry	-	-	160
			wet	-	-	120
Tensile strength Warp	EN2561 type B	MPa	dry	-55	475	435
				RT	530	475
				90	530	475
			wet	70	530	475
				90	530	475
Tensile modulus Warp	EN2561 type B	GPa	dry	-55	54 ± 7	
				RT	54 ± 7	
				70	54 ± 7	
			wet	70	54 ± 7	
				90	54 ± 7	
(continued)						

(continued)

Table 5: Mechanical properties

Properties	Test method	Unit	Conditioning ²⁾	Test temp. [°C]	Requirements ¹⁾	
					X	Xmin
Material Code					AIMS05-01-005A	
Tensile strength Weft	EN2597 type B	MPa	dry	-55	490	440
				RT	500	475
			wet	70	490	475
				90	490	475
Tensile Modulus Weft	EN2597 type B	GPa	dry	-55	54 ± 7	
				RT	54 ± 7	
			wet	70	54 ± 7	
				90	54 ± 7	
Compression strength Warp	EN2850 type B	MPa	dry	RT	550	500
				90	440	400
			wet	70	440	405
				90	430	390
Compression modulus Warp	EN2850 type B	GPa	dry	-55	47 ± 6	
				RT	47 ± 6	
				90	47 ± 6	
			wet	70	47 ± 6	
				90	47 ± 6	
Compression strength Weft	EN2850 type B	MPa	dry	RT	550	500
			wet	70	425	405
				90	340	300
Compression modulus Weft	EN2850 type B	GPa	dry	RT	47 ± 6	
			wet	70	47 ± 6	
				90	47 ± 6	
In plane shear strength	AITM1-0002	MPa	dry	-55	110	
				RT	110	
				90	80	
				120	65	
			wet	RT	85	
				70	80	
				90	70	
				120	55	
(continued)						

(continued)

Table 5: Mechanical Properties

Properties		Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1)	
						X	Xmin
Material Code						AIMS 05-01-005A	
In plane shear modulus		AITM1-0002	GPa	dry	-55	4,7 ± 0,7	
					RT	3,5 ± 0,7	
					90	2,7 ± 0,5	
					120	2,2 ± 0,5	
				wet	RT	3 ± 0,5	
					70	2,5 ± 0,5	
					90	2,2 ± 0,5	
					120	1,1 ± 0,3	
Compression after impact 3) 4)		AITM1-0010	MPa	dry	RT	180	160
				wet	90	155	135
G ₁ C		AITM1-0005	J/m ²	dry	RT	210	190
				wet	RT	305	280
G ₂ C		AITM1-0006	J/m ²	dry	RT	2600	2400
Multiangular Tension	Unnotched 4)	AITM1-0007 type A (Quasi-isotropic)	MPa	dry	-55	350	320
					RT	350	320
					90	350	320
	wet	90		350	320		
	Notched 4) 5)	AITM1-0007 type B (Quasi-isotropic)		dry	-55	180	165
					RT	180	165
				wet	90	180	165
	Notched (filled) 4) 5)	AITM1-0007 type C (Quasi-isotropic)		dry	RT	180	165
					90	180	165
				wet	90	180	165
90			180		165		
(continued)							

(continued)

Table 5: Mechanical Properties

Properties		Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1)	
						X	Xmin
Material Code						AIMS05-01-005A	
Multiangular Compression	Unnotched 4)	AITM1-0008 type A (Quasi- isotropic)	MPa	dry	RT	380	350
					90	325	290
				wet	90	215	200
	Notched 4) 5)	AITM1-0008 type B (Quasi- isotropic)		dry	-55	270	245
					RT	225	205
				wet	90	170	155
					Notched (filled) 4) 5)	AITM1-0008 type C (Quasi- isotropic)	dry
	90	230		210			
	wet	90		200			180
		water		90			200
Bearing Strength 4)	2%	AITM1-0009 type A (Quasi- isotropic)	MPa	dry			-55
					RT	600	550
				wet	90	400	350
					120	300	260
	Ultimate	AITM1-0009 type A (Quasi- isotropic)	MPa	dry	-55	680	630
					RT	680	630
				wet	90	475	425
					120	350	310
Interlaminar Shear Strength (warp)		EN2563	MPa	dry	-55	51	45
					RT	60	55
					90	50	45
					120	42	38
				wet	RT	50	45
					70	47	42
					90	37	32
					120	32	27
				water	90	32	27
				(continued)			

(continued)

Table 5: Mechanical properties (concluded)

Properties	Test method	Unit	Conditioning 2)	Test temp. [°C]	Requirements 1)	
					X	Xmin
Material Code					AIMS05-01-005A	
Resistance to aggressive fluids 6)						
Reference	EN2563	MPa	dry	RT	51	45
			wet	RT	50	45
			wet	90	37	32
			water	90	32	27
Skydrol			70°C / 1000h	90	8)	8)
Skydrol/water			50°C / 1000h	90	8)	8)
Fuel			RT/1000h	90	8)	8)
MEK			RT/1h	RT	8)	8)
Paint Strippers 7) (Non-phenolic)	AITM1-0001	MPa	dry	RT	9)	
Sandwich properties						
Flatwise Tensile Strength	AITM1-0025	MPa	dry	-55	7,0	6,0
				RT	7,0	6,0
				120	9)	
			wet	70	9)	
Fluid tightness test	AITM2-0037	MPa	-	-	Pass 10)	
Sandwich flexure 11)	AITM1-0018	N	dry	RT	1300	1200
Flatwise Tensile Strength	AITM1-0025	MPa	dry	-55	7,0	6,0
				RT	7,0	6,0
				120	9)	
			wet	70	9)	
Fluid tightness test	AITM2-0037	MPa	-	-	Pass 10)	
Sandwich flexure 11)	AITM1-0018	N	dry	RT	1300	1200

Notes:

- 1) - X means minimum average value.
- X_{min} means minimum individual value.
- Requirements for properties: Tensile and compression strength (warp), Plain Tension and compression, Open hole tension and compression, and Filled hole tension and compression are calculated with a nominal cured ply thickness of 0,237 mm, fiber volume = 46,3% (see Table 4). Other requirements are calculated with real thickness.
- 2) Conditioning: "dry" according to EN2743; "wet" is 70°C/85% relative humidity according to EN2823, "water" is 70°C/water according to EN2378.
- 3) Impact energy 3,3 J/mm.
- 4) Lay up (+0/-90)_{2s}, quasi-isotropic.
- 5) Requirements calculated with gross section.
- 6) According to AIMS05-01-000 Part 1.
- 7) According to AIMS05-01-000 Part 1 or according to QTP.
- 8) Values after conditioning will have no significant decrease compared to the reference value tested at the same temperature.
- 9) Results to be reported during qualification.
- 10) This test will be carried out during manufacturing trials for qualification.
- 11) Cocured Specimens Lay up: 0°/90°/IPS10-01-006-02 Grade B/Glass honeycomb 3.0-3/16, 12,7 mm thickness/IPS10-01-006-02 Grade B/90°/0°.

5 Designation

Refer to ABS5346.

6 Technical Specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-01-000 Part 1 and Part 6.

7 Incoming Inspection

Material qualified against this AIMS shall be subject to "Technical Incoming Inspection" as per AIMS05-01-000 Part 100 prior to use.

Record of Revisions

Issue	Clause modified	Description of Modification
1 02.2006		New Standard
2 09.2011	Table 5	Change of requirements for several conditions for: Tensile and Compression strength (warp and weft) – minimum individual; Compression modulus (warp and weft – tolerance; IPS strength and ILSS – minimum individual / average
3 07.2013	3 7	AIMS05-01-000 Part 100 added Incoming Inspection added
4 09.2018	All Table 5	Updated to latest template and reformatted as required Change of requirements for IPSM dry; G ₂ C wet requirement deleted